



UIT

UNIVERSITY OF INFORMATION TECHNOLOGY & SCIENCES

Assignment

Course Code-CSE412

Course Title-Operating System Lab

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```

#include <stdio.h>
#include <limits.h>

int main() {
    int n, i, smallest, time = 0, completed = 0;
    int p[10], at[10], bt[10], remaining_bt[10], ct[10], tat[10], wt[10];
    float total_tat = 0, total_wt = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter process IDs: ");
    for (i = 0; i < n; i++) {
        scanf("%d", &p[i]);
    }

    printf("Enter arrival times: ");
    for (i = 0; i < n; i++) {
        scanf("%d", &at[i]);
    }

    printf("Enter burst times: ");
    for (i = 0; i < n; i++) {
        scanf("%d", &bt[i]);
        remaining_bt[i] = bt[i];
    }

    while (completed < n) {
        smallest = -1;
        int min_bt = INT_MAX;

        for (i = 0; i < n; i++) {
            if (at[i] <= time && remaining_bt[i] > 0 && remaining_bt[i] < min_bt) {
                min_bt = remaining_bt[i];
                smallest = i;
            }
        }

        if (smallest == -1) {
            time++;
            continue;
        }

        remaining_bt[smallest]--;
        time++;

        if (remaining_bt[smallest] == 0) {
            completed++;
            ct[smallest] = time;
            tat[smallest] = ct[smallest] - at[smallest];
            wt[smallest] = tat[smallest] - bt[smallest];
            total_tat += tat[smallest];
            total_wt += wt[smallest];
        }
    }

    printf("\nP\tAT\tBT\tCT\tTAT\tWT");
    for (i = 0; i < n; i++) {
        printf("\nP%d\t%d\t%d\t%d\t%d\t%d", p[i], at[i], bt[i], ct[i], tat[i], wt[i]);
    }

    printf("\n\nAverage Turnaround Time: %.2f", total_tat / n);
    printf("\nAverage Waiting Time: %.2f\n", total_wt / n);

    return 0;
}

```

main.c

Output



```

Enter number of processes: 5
Enter process IDs: 1 2 3 4 5
Enter arrival times: 0 1 2 4 6
Enter burst times: 5 3 6 2 4

```

P	AT	BT	CT	TAT	WT
P1	0	5	10	10	5
P2	1	3	4	3	0
P3	2	6	20	18	12
P4	4	2	6	2	0
P5	6	4	14	8	4

```

Average Turnaround Time: 8.20
Average Waiting Time: 4.20

```

=== Code Execution Successful ===

Output

```

Enter number of processes: 4
Enter process IDs: 1 2 3 4
Enter arrival times: 0 1 2 3
Enter burst times: 8 4 9 5

```

P	AT	BT	CT	TAT	WT
P1	0	8	17	17	9
P2	1	4	5	4	0
P3	2	9	26	24	15
P4	3	5	10	7	2

```

Average Turnaround Time: 13.00
Average Waiting Time: 6.50

```